APÉRY'S CONSTANT ON ONE PAGE

# $\zeta(3)$ in place of $\zeta(2)$

The sum of reciprocal cubes, and what happens to Ramanujan's constant when $\pi$ is rewritten through it.

THE RULE

## The sum of reciprocal cubes, and Apéry's faster series

$$\zeta(3) \;=\; \sum_{n\ge 1}\frac{1}{n^3} \;=\; \frac{5}{2}\sum_{n\ge 1}\frac{(-1)^{n-1}}{n^3\binom{2n}{n}} \;=\; 1.202\,056\,903\,159\,594\ldots$$

Five terms of the plain sum leave an error of $0.016$, while the central binomial coefficient $\binom{2n}{n}$, roughly quadrupling at each step, makes Apéry's series gain about $0.6$ of a digit per term. Roger Apéry used series of this kind in 1978 to prove that $\zeta(3)$ is irrational, and Beukers later recast the proof through the integral

$$\zeta(3) = -\frac{1}{2}\int_0^1\!\!\int_0^1 \frac{\ln(xy)}{1-xy}\,\mathrm{d}x\,\mathrm{d}y.$$

| $n$ | $\sum 1/k^3$ | Apéry's sum |
|---|---|---|
| 1 | 1 | 1.25 |
| 2 | 1.125 | 1.197 9 |
| 3 | 1.162 0 | 1.202 55 |
| 4 | 1.177 7 | 1.201 99 |
| 5 | 1.185 7 | 1.202 068 |
| $\infty$ | | 1.202 056 9 … |

Five terms of each: the plain sum is still in the second decimal place.

EVEN VALUES ARE EASY

## Odd values are hard

Euler showed that every even zeta value is a rational multiple of a power of $\pi$,

$$\zeta(2)=\frac{\pi^2}{6},\quad \zeta(4)=\frac{\pi^4}{90},\quad \zeta(6)=\frac{\pi^6}{945},$$

but no such form is known for any odd value. Nobody knows whether $\zeta(3)/\pi^3 \approx 0.038\,77$ is rational, whether $\zeta(5)$ is irrational, or whether $\zeta(3)$ is transcendental; it is known that infinitely many odd values are irrational and that at least one of $\zeta(5),\zeta(7),\zeta(9),\zeta(11)$ is.

WHERE IT APPEARS

## Coprime triples and black-body light

Three whole numbers chosen at random share no common prime factor with probability $1/\zeta(3) \approx 0.832$, the cube analogue of $6/\pi^2$ for pairs. In physics, a black body at temperature $T$ holds

$$n_\gamma = \frac{2\,\zeta(3)}{\pi^2}\left(\frac{kT}{\hbar c}\right)^3$$

photons per unit volume, which is about $411$ per cubic centimetre in the cosmic microwave background; $\zeta(3)$ also appears in the electron's magnetic moment.

THE EXPERIMENT

## Rewrite $\pi$ through $\zeta(3)$ and leave $163$ alone

In $\mathcal{R} = \mathrm{e}^{\sqrt{978\,\zeta(2)}}$ the factor $978$ is $6\times 163$, and the $6$ belongs to $\zeta(2)$ through $\pi^2 = 6\,\zeta(2)$. The faithful rewrite for $\zeta(3)$ uses $\pi^3 = \kappa_3\,\zeta(3)$ with $\kappa_3 = \pi^3/\zeta(3)$, which changes the formula without changing the number:

$$\mathcal{R} \;=\; \mathrm{e}^{\sqrt{163}\,(\kappa_3\,\zeta(3))^{1/3}}, \qquad \kappa_3 \;=\; 25.794\,350\,166\,618\,684\ldots$$

So $640320^3$ survives untouched, and the short form now runs backwards to give $\kappa_3 \approx \left[\ln(640320^3+744)\right]^3/\left(163^{3/2}\zeta(3)\right)$, correct to 29 decimal places. The table shows the reverse dependence: each further digit of $\kappa_3$ pins down more of $640320$.

| $\kappa_3$ used | $\sqrt[3]{\mathcal{R}-744}$ |
|---|---|
| 25.79 | 639 838.89 |
| 25.794 | 640 281.26 |
| 25.794 35 | 640 319.98 |
| exact | 640 320 |

Truncating $\kappa_3$ rebuilds a slightly wrong $640320$.

WHY $\zeta(2)$ GAVE A WHOLE NUMBER

## Even $\kappa_s$ are rational, odd ones are unexplained

For every even $s$ Euler's formula makes $\kappa_s = \pi^s/\zeta(s)$ rational, so Ramanujan's constant has a family of exact rewrites,

$$\mathcal{R} = \mathrm{e}^{\sqrt{163}\,(6\,\zeta(2))^{1/2}} = \mathrm{e}^{\sqrt{163}\,(90\,\zeta(4))^{1/4}} = \mathrm{e}^{\sqrt{163}\,(945\,\zeta(6))^{1/6}},$$

each as clean as the original. For odd $s$ nobody knows whether $\kappa_s$ is rational; if $\kappa_3$ were, $\zeta(3)$ would be a rational multiple of $\pi^3$, and nothing points that way. Ramanujan's constant can therefore carry $\zeta(3)$ only numerically, through the unexplained $25.794\ldots$, while the integer $640320^3$ is fixed by $\pi$ and $163$ alone.

| $s$ | $\kappa_s = \pi^s/\zeta(s)$ | |
|---|---|---|
| 2 | 6 | rational |
| 3 | 25.794 350 … | unknown |
| 4 | 90 | rational |
| 5 | 295.121 509 … | unknown |
| 6 | 945 | rational |
| 7 | 2995.284 764 … | unknown |

The factor that converts $\zeta(s)$ into a power of $\pi$.

**Conventions.** $\kappa_s = \pi^s/\zeta(s)$, so that $\pi = (\kappa_s\zeta(s))^{1/s}$ for every $s \ge 2$; all figures were computed to 50 digits.